Functions and Function Notation

Section 1: Evaluating Functions and Function Notation

1 Given $f(x) = 3x^2 - 4x + 1$, evaluate $f(-2)$.

a. 21

b. 5

c. -3

d. 17

2 If $g(x) = \frac{2x+5}{x-1}$, find $g(3a+1)$.

a. $\frac{6a+7}{3a}$

b. $\frac{6a+5}{3a+1}$

c. $\frac{6a+2}{3a}$

d. $\frac{2a+5}{a}$

3 Find all values of x for which $f(x) = x^2 - 6x$ equals -8.

a. $x = 2$ and $x = 4$

b. $x = -2$ and $x = -4$

c. $x = 1$ and $x = 8$

d. $x = 0$ and $x = 6$

Section 2: The Vertical Line Test and Relations

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4 Which of the following equations defines y as a function of x?

a. $y = 2x^3 - 5$

b. $x^2 + y^2 = 16$

c. $x = y^2 + 1$

d. $|y| = 3x$

5 What does the Vertical Line Test determine about a graph?

a. Whether the graph represents a function

b. Whether the function is one-to-one

c. The domain of the function

d. The slope of the curve

6 Which set of ordered pairs represents a function?

a. $\{(1, 2), (2, 3), (3, 4), (4, 4)\}$

b. $\{(1, 2), (1, 3), (2, 4)\}$

c. $\{(0, 5), (0, -5), (1, 2)\}$

d. $\{(3, 1), (3, 2), (3, 3)\}$

Section 3: Operations and Difference Quotients

7 Simplify the difference quotient $\frac{f(x+h) - f(x)}{h}$ for $f(x) = 4x - 7$.

a. 4

b. $4h$

c. $4x + 4h - 7$

d. 0

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8 Simplify $\frac{f(x+h)-f(x)}{h}$ for $f(x) = 2x^2 + 3$.

a. $4x + 2h$

b. $4x$

c. $2x + 2h$

d. $4x + h$

9 If $f(x) = x^2 - 1$ and $g(x) = 2x + 3$, find $(f \cdot g)(2)$.

a. 21

b. 15

c. 18

d. 28

Section 4: Function Composition

10 Given $f(x) = 2x + 1$ and $g(x) = x^2 - 3$, find $(f \circ g)(x)$.

a. $2x^2 - 5$

b. $4x^2 + 4x - 2$

c. $2x^2 - 3$

d. $2x^2 + 1$

11 For the same functions $f(x) = 2x + 1$ and $g(x) = x^2 - 3$, evaluate $(g \circ f)(2)$.

a. 22

b. 25

c. 17

d. 3

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12 If $h(x) = \sqrt{2x^3 - 5}$, find functions $f(x)$ and $g(x)$ such that $h(x) = (f \circ g)(x)$.

a. $f(x) = \sqrt{x}, g(x) = 2x^3 - 5$

b. $f(x) = 2x^3 - 5, g(x) = \sqrt{x}$

c. $f(x) = x^3, g(x) = \sqrt{2x - 5}$

d. $f(x) = 2\sqrt{x}, g(x) = x^3 - 5$

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